Amir Daghighi

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AI-native computational chemist (**5+ years**) with strong statistical and mathematical background. Leveraged Python and deep learning methods at NDSU to develop predictive ADMET models. Proven track record of enhancing computational models' performance at previous internships at **SafetySpect Inc** and **OpenEye Scientific**. Adept in Python, with strong problem-solving abilities and a history of achieving high-impact results.

EDUCATION

PhD in Biomedical Engineering **May 2023 – Present**
North Dakota State University, Fargo, ND *Expected Graduation: May 2027*
GPA 4.00/4.00

MSc in Biomedical Engineering **Jan 2021 – May 2023**
North Dakota State University, Fargo, ND
GPA 4.00/4.00

Bachelor of Science, Biomedical Engineering (Bio-electrical Eng) **Sep 2016 – Dec 2020**
Minor: Biomedical Image & Signal Processing
Amirkabir University of Technology (Tehran Polytechnique), Tehran, Iran
GPA 3.85/4.00

TECHNICAL SKILLS

Programming Languages & Version Control: Python, C#, C/ C++, SQL, Shell scripting, BASH, Git
ML toolkit & workflow: TensorFlow, Keras, PyTorch, PyG, scikit-learn, LlamaIndex, LangChain, Streamlit, Pandas
Operating Systems: Linux, Ubuntu, MacOS, Windows
Deep Learning Specialties: NLP, LLM, RNN, LSTM, MLP, CNN, GNN, GraphSAGE
Drug Discovery: VASP, GROMACS, MOE, Schrodinger, AI-based molecule design, RDKit, Gaussian, VMD, OpenBabel
Theoretical skills: Statistics, Linear Algebra, Graph Theory, Calculus, Algorithm development, OOP, problem solving, critical thinking
Related coursework: Computational Quantum Chem, Computational Material Design, Advanced Data Mining, Graph-Based Data Analytics, Neural Networks, Numerical Analysis, Statistics

PROFESSIONAL EXPERIENCE

Graduate Research Assistant, NDSU, Fargo, ND **Jan 2021 – Present**

- Deep Learning modeling for toxicity prediction across 80,000+ compounds and 59 biological endpoints.
- Utilizing High-performance Cluster (HPC) for training, hyper-parameter tuning, and interpretation of QSAR models.
- Developed agentic AI workflows for automated QSAR model development using LangGraph.

Scientific Developer Intern, OpenEye Cadence Molecular Sciences, Santa Fe, NM **May 2024 – Aug 2024**

- Developed a confidence-weighted ensemble algorithm combining 2D and 3D QSAR models.
- Established 3D pose-quality thresholds to optimize model selection and improve prediction accuracy.
- Benchmarked models across 23 curated protein datasets using statistical t-tests, demonstrating comparable or superior performance to third-party industry models.

Machine Learning Engineer Intern, SafetySpect Inc, Grand Forks, ND **Jun 2022 – Sep 2022**

- Developed a novel hierarchical ML classification framework for identifying 43 fish species using multimodal spectroscopy data.
- Implemented neural network models (MLP) and data fusion techniques integrating fluorescence, VNIR, and SWIR spectroscopy.
- Improved classification accuracy by up to 40% for individual species, with multimodal fusion delivering a 9% improvement over the best single-mode model.

AWARDS AND HONORS

- NDSU College of Arts and Sciences Graduate Research Award Spring 2026
- Travel award at ASCCT conference 14th annual meeting, Washington, DC Fall 2025
- Excellent presentation award, Red River Valley Local ACS conference, Grand Forks, ND Spring 2025
- 3-Minute Thesis Innovation Commendation award at NDSU 3MT competition Spring 2024
- Travel award at ASCCT conference 12th annual meeting, Silver Spring, MD Fall 2023
- Tox21 Student Award at ASCCT conference 12th annual meeting, Silver Spring, MD Fall 2023
- Outstanding oral presentation award, Red River Valley Local ACS conference, Moorhead, MN Spring 2023
- Active member of ACS conference (attended meetings in Washington DC, 2025 – New Orleans, LA, Spring 2024 – Chicago, IL, summer 2022 – Atlanta, GA, Spring 2021)

PUBLICATIONS

- https://scholar.google.com/citations?hl=en&user=3WBjqm0AAAAJ